DeepKids Mission Book · Plants: parts and kinds · Core

The Case of the Drooping Tulsi

A story told by a tree, a real plant mystery, and the biggest ideas about plants: wood made of air, plant minds, weeds, and seeds that belong to everyone.

Detective's name: _____ Started on: _____

🔎 The case

Dadi's tulsi has always been the happiest plant in the house. But this month it is drooping. Its leaves are turning yellow and dropping off, and the soil smells a little like old socks.

Dadi is worried. She looks after it more than ever: every single morning she pours a full lota of water into its beautiful new painted pot. And last month she moved it from the sunny balcony to the kitchen corner, so she can see it while she cooks.

Something is wrong, and you are going to find out what. But first, a story.



Your mission: Read the story. Visit all 10 stations and collect the clues. At the end, write your case report and save the tulsi.

Your stations One a day is plenty. Tick each one when it's done.

Station 1	How a plant works	≈ 35 min ☐
Station 2	Lab: Does water really travel up?	≈ 40 min ☐
Station 3	A tree made of air	≈ 40 min ☐
Station 4	Survey: who decides what's a weed?	≈ 40 min ☐
Station 5	Leaf and root detectives	≈ 40 min ☐
Station 6	Are plants clever?	≈ 40 min ☐
Station 7	Seeds: travelling and shared	≈ 45 min ☐
Station 8	Puzzle room	≈ 30 min ☐
Station 9	Solve the case	≈ 40 min ☐
Station 10	Workshop: make, grow and test	≈ 60 min ☐

You'll need (ask a grown-up)

- ☐ a clear plastic bag
- ☐ haldi, a beetroot and onion skins
- ☐ a glass and red ink or food colour
- ☐ a touch-me-not (lajwanti) plant if you can find one
- ☐ moong and cotton
- ☐ clay, compost and seeds for seed balls
- ☐ a white flower, a balsam (gulmehendi) stem or a celery stick
- ☐ a hand lens if you have one

- ☐ 6 different leaves
- ☐ crayons and a pencil
- ☐ a small weed or a coriander plant you may pull up
- ☐ black paper and paper clips
- ☐ scissors and a paper clip
- ☐ a stopwatch or phone

About 7 hours, over two weeks in all

Story time

A STORY TO READ ALOUD

The Neem Who Was Made of Breath

I am a neem tree, and I have stood at the corner of Kumbhar Lane for eighty-one years. I do not have eyes, but I know everyone who lives here. I know them by their breath.

You think that is strange. Let me explain. Every time you breathe out, a little puff of invisible gas leaves your body. You call it carbon dioxide. You think it is nothing, just old air. But to me it is food. I pull it in through millions of tiny mouths on the undersides of my leaves, and with the sunlight that falls on me, and the water my roots drink from deep under the lane, I turn it into sugar, and the sugar into wood, and leaves, and the little bitter fruits the parrots love.

So I am made of your breath. My trunk, my branches, every one of my leaves: they are built mostly out of the air. Out of the breath of everyone who ever sat beneath me.

I remember Shanta when she was a baby, eighty years ago, sleeping in a cloth cradle tied to my lowest branch while her mother pounded rice. I breathed in her baby breath, and some of it became the ring of wood I grew that year. Shanta grew up, and grew old, and now she is the grandmother of the lane, and she still sits under me every evening. Her breath has changed. It is slower now. But I know it.

And here is the other half of the secret, the half that you forget. When I make my food, I breathe out too. I breathe out the gas you call oxygen, and you breathe it in. So Shanta and I have been breathing each other for eighty years. Her out-breath is my in-breath. My out-breath is hers. We are not two separate things, the old woman and the tree. We are one long, slow breath, going back and forth.

This summer, men came with measuring tapes. The road was to be widened, they said, for the trucks. The tree at the corner would have to go.

"What is it worth?" asked the man with the clipboard. "The timber, I mean. Neem wood is good wood. Five thousand rupees, perhaps."

Shanta's granddaughter, Gauri, was nine. She had been listening. "Five thousand rupees?" she said. "Then who will pay for the shade? In May it is ten degrees cooler under the neem than on the road. Who will pay for that?"

The man laughed. But Gauri kept going. She had a notebook. "Who will pay the parrots, and the bulbuls, and the squirrels who live in it? Who will pay for the oxygen? Who will pay my Aaji for eighty years of shade? You can only put a price on the wood because the wood is the only part you can sell."

The man stopped laughing.

The next morning, when the men came back with their saws, they found the women of Kumbhar Lane standing around my trunk. Shanta, with her walking stick. Gauri's mother, who sells vegetables. The washerwoman, the potter's wife, the girls from the school. They held hands around me, the way the women of Reni village in the Himalayas once held hands around their trees, long ago, and would not let them be cut.

"This tree is made of our breath," Gauri told the men. "My grandmother's breath, and my mother's, and mine. If you cut it, you cut that."

I do not know if the men understood about the breath. Perhaps it was the women standing so quietly. Perhaps it was the shade, which was very pleasant that

hot morning, and made them think. They went away. They made the road go round the corner instead of through it.

That evening, Shanta sat under me as she always does. Gauri sat beside her. They were quiet, breathing, and I was quiet, breathing.

Somewhere in me, a new ring of wood is growing this year. It is made of the air of that morning: the breath of the women who held hands, and the breath of the men who went away. I will keep it for a hundred years.

Talk about it

1. Where does the wood of a tree come from? Were you surprised?
2. Why does the tree say it is 'made of your breath'?
3. What do the old woman and the tree give each other?
4. The man said the tree was worth ₹5,000. What did Gauri mean when she said he could only price the wood?
5. Why do you think the women held hands around the tree?

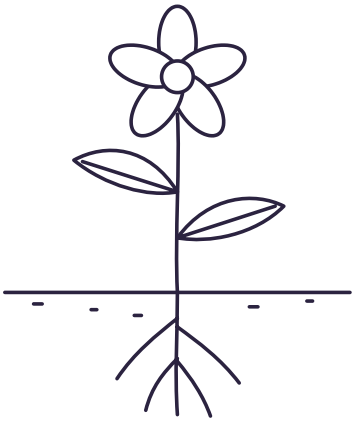
Station 1 How a plant works

≈ 35 min ☐ done

A plant is a food factory

We get our food from the kitchen. A plant makes its own food, right inside its leaves, and it needs only three things to do it: sunlight, water and air.

The roots hold the plant firmly and soak up water and minerals from the soil. The stem holds the leaves up to the light and carries water up and food down. The flowers make seeds.



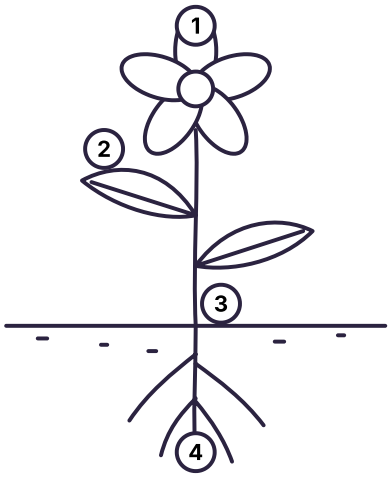
Every part has a job.

The leaves are the kitchens. A green substance in them, chlorophyll, catches sunlight. Using that energy, the leaf joins water (from the roots) with carbon dioxide (from the air) to make sugar, the plant's food. As it does this, it gives out oxygen, the gas we breathe. Scientists call this photosynthesis: 'making with light'.

Think about what that means. Every bite of food you have ever eaten, rice, dal, mango, even milk and eggs, began as sunlight caught by a leaf. You are, in a way, made of sunshine.

2 Label the plant

Write the words from the box next to the numbers.



leaf **stem** **root** **flower**

1. _____

2. _____

3. _____

4. _____

Now write what each part does:

1 does this: _____

2 does this: _____

3 does this: _____

4 does this: _____

🗝️ **Clue 1** Leaves make the plant's food, but only when they get enough sunlight.
Copy it onto the Case board on the last pages.

3 Mini lab: the covered leaf

Question: Does a leaf need light to stay healthy and green?
You need: a potted plant with big leaves (not the tulsi!), black paper, paper clips

1. Fold black paper over half of one leaf, front and back. Clip it on gently.
2. Leave the plant in its usual sunny place for 4 days. Water it as normal.
3. On day 4, take off the paper and compare the covered half with the uncovered half.

I predict:

_____

_____

because

_____



day 1 | day 4: covered half | day 4: uncovered half

What I found out

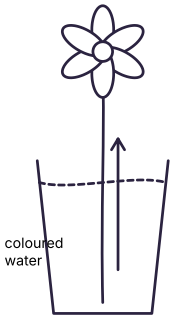
The covered part looked _____

I think this is because _____

Station 2 **Lab: Does water really travel up?** ≈ 40 min ☐ done

4 The coloured-water test

Question: If a plant drinks through its roots, how does water get up to the leaves and flowers?
You need: a glass, red ink or food colour, a white flower, a balsam stem or a celery stick, scissors (a grown-up cuts)



1. Fill the glass halfway with water and stir in lots of colour, until it's dark.
2. Ask a grown-up to cut the bottom of the stem at a slant, under running water.
3. Stand the stem in the coloured water. Note the time.
4. Look after 1 hour, 6 hours and the next morning. Draw what you see each time.
5. Next day: ask a grown-up to cut across the stem. Look at the cut end with a hand lens.

I predict:

_____

_____

because

_____

Time	Where is the colour?	How far up (cm)?
start		0
1 hour		
6 hours		
next day		

after 1 hour	after 6 hours	next morning

What I found out

The colour went _____

This shows that water travels _____

In the cut stem I saw _____

A mystery about water

A tall tree like a eucalyptus or a redwood lifts water more than 100 metres, as high as a 30-storey building, with no pump and no heart. How? Scientists' best answer: as water evaporates from the leaves, it pulls the water below it up, like a long, thin, unbroken rope of water. But water 'ropes' should snap easily. How the tubes keep from breaking, and how a tree repairs them when air bubbles get in, is still being studied.

🗝️ **Clue 2** Water moves up from the roots. But roots also need AIR: in soil that is always soaked, there is no air, and the roots start to rot and smell.
Copy it onto the Case board on the last pages.

Station 3 A tree made of air

≈ 40 min ☐ done

The willow puzzle

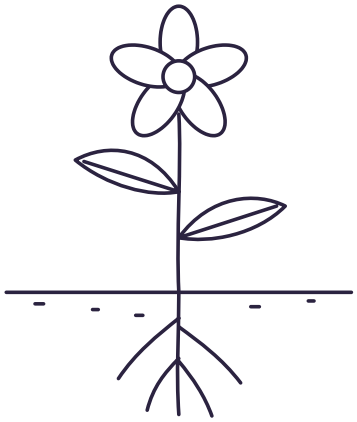
About 400 years ago, a scientist in Belgium called Jan Baptist van Helmont wondered where a tree's wood comes from. Most people thought it was obvious: trees eat soil.

So he tested it. He dried 200 pounds of soil and put it in a big pot. He planted a little willow tree weighing 5 pounds. For five years he gave it nothing but rainwater, and covered the pot so no dust could get in. Then he took the tree out and weighed it: about 169 pounds! He dried the soil and weighed it again. It had lost only about 2 ounces, the weight of a small mango.

So the tree had gained about 164 pounds, and the soil had lost almost nothing. Where did the tree come from?

5 Be the scientist

Van Helmont decided the wood came from the water. He was partly right, but he missed something invisible. Think it through.



1. The tree gained 164 pounds. The soil lost about 2 ounces. Could the wood have come from the soil? Why not?

2. Van Helmont said: from the water. What else was around the tree all the time that he couldn't see or weigh?

3. Today we know most of the wood's weight comes from carbon dioxide in the air. So when a tree burns, where does most of it go?

The breath we share

Here is one of the most beautiful ideas in all of science. When you breathe out, you give carbon dioxide to the air. Plants take it in and build themselves out of it. When plants make food, they give out oxygen, and you breathe it in. Some of the air in your lungs right now came from a plant. Some of the wood in the neem outside may once have been your breath.

The carbon in your body came from the food you ate, which came from plants, which took it from the air. Before that, long ago, it was made inside stars. You, the tree and the stars are made of the same stuff, passed around and around.

6 Breath count

Question: How much do we depend on plants, every minute?
You need: a stopwatch, a quiet place, ideally under a tree

1. Sit still under a tree for one minute. Count your breaths.
2. Count again after running for one minute.
3. Every breath in brings oxygen that plants (on land and in the sea) put into the air. Fill in the table.

I predict:

_____

_____

because

_____

	Breaths in 1 minute	In 1 hour (× 60)	In a day (× 24)
sitting			
after running			

What I found out

In a day I take about _____ breaths, and every one depends on

Something I want to say to a tree:

Station 4 Survey: who decides what's a weed?

≈ 40 min ☐ done

Five kinds of plants, and one word to question

Trees are big with one thick, woody trunk (neem, mango, peepal). Shrubs are woody but smaller, with many stems near the ground (hibiscus, rose). Herbs are small with soft green stems (coriander, mint). Climbers need a support (money plant, bitter gourd). Creepers spread along the ground (pumpkin). Grasses have long thin leaves (doob, wheat, rice, and bamboo, which is a giant grass).

Now a harder question. What is a weed? A gardener will say: a plant growing where we don't want it. Doob grass is a weed in a vegetable patch, but it is sacred in a puja. Is the plant different, or only our wish? A 'weed' in a crack in the pavement might be feeding bees, holding soil, or cooling a tiny patch of ground. Some plants really do harm, like the invasive congress grass (Parthenium), which can push out other plants and make people itch. But who gets to decide what a plant is for?

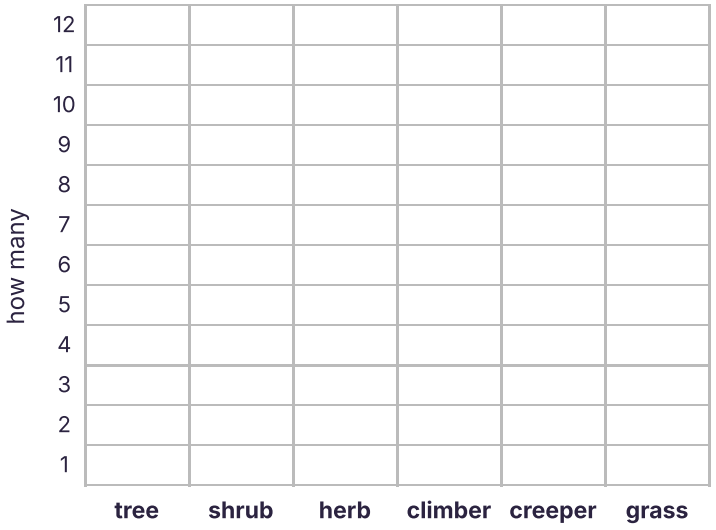
7 Plant count

Walk around your street, building or a park with a grown-up for 20 minutes. Tally every plant you see. Put a ★ next to any that people would call weeds.

Kind	Tally (𝍸)	Total	One example I found
tree			
shrub			
herb			
climber			
creeper			
grass			

8 Make a bar graph

Colour one square for each plant. If you counted more than 12, let each square stand for 2.



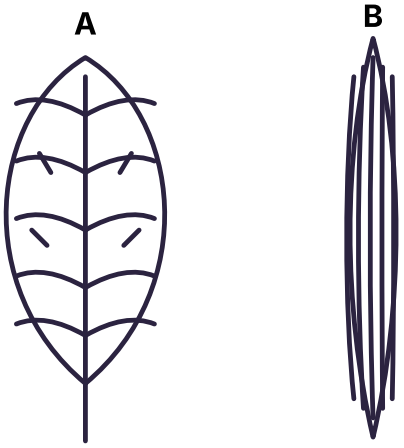
1. Which kind did you see most? Why do you think so?

2. Pick one 'weed' you found. Watch it for 5 minutes. Who visits it or lives on it? What is it doing for them?

Station 5 **Leaf and root detectives** ≈ 40 min ☐ done

9 Two kinds of leaf veins

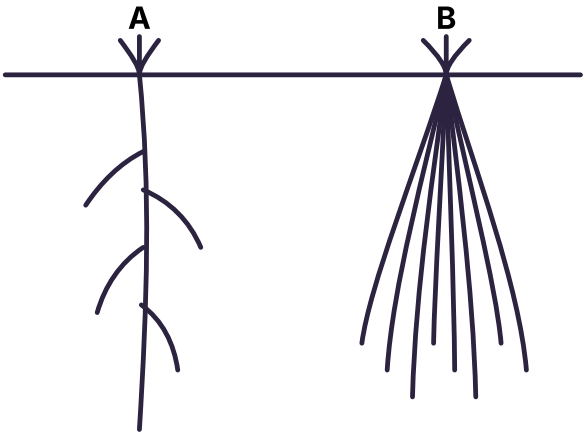
Hold a leaf up to the light: you can see its veins, the pipes that carry water. Leaf A has a net of veins. Leaf B has veins running side by side (parallel).



1. Collect 6 different leaves. Rub over each with a crayon under thin paper. Is each one A (net) or B (parallel)?

10 Two kinds of roots

Root A has one thick main root with thin ones branching off: a tap root. Root B is a bunch of thin roots all the same size: fibrous roots.



11 Find the secret rule

Question: Is there a link between a plant's leaf veins and its kind of root?
You need: a weed or grass pulled up from a crack, a coriander or mustard plant pulled up from a pot (with a grown-up's permission), a hand lens

1. Pull up a grass plant gently. Look at its leaves and its roots.
2. Do the same with a coriander or mustard plant.
3. Fill in the table. Some rows are done for you.
4. Look for the pattern.

I predict:

_____

_____

because

_____

Plant	Leaf veins: net or parallel?	Roots: tap or fibrous?
grass (yours)		
coriander (yours)		
wheat	parallel	fibrous
bean	net	tap
maize	parallel	fibrous
mustard	net	tap

What I found out

The secret rule is: net veins usually go with _____

and parallel veins usually go with _____

A banana leaf has parallel veins. I predict its roots are _____

Station 6 **Are plants clever?** ≈ 40 min ☐ done

A plant that learns?

Plants have no brain and no nerves. And yet they sense light, gravity, touch, water, and even the chemicals other plants give off. The touch-me-not (lajwanti, Mimosa pudica) folds its leaves in a second when touched.

In 2014, a scientist named Monica Gagliano tested whether a plant could learn. She dropped potted touch-me-nots a short, harmless distance again and again. At first they folded their leaves every time. After a while, they stopped folding: as if they had learned the drop was nothing to fear. When she shook them in a new way, they folded again, so they weren't just tired. She said they remembered for weeks.

Some scientists were excited: maybe plants can learn without a brain! Others said: careful. 'Learning' and 'intelligence' are words we made for animals with brains. Maybe something simpler is going on. The argument is still going on today.

12 Touch-me-not lab

Question: Does a touch-me-not get used to being touched?
You need: a touch-me-not plant, a stopwatch, a soft paintbrush or feather

1. Touch one leaf gently with the brush. Time how long until it opens again.
2. Choose another leaf. Touch it once every minute for 10 minutes. Does it fold less as time goes on?
3. Now touch a leaf that hasn't been touched today. What happens?
4. Be gentle: the plant spends energy each time it folds.

I predict:

_____

_____

because

_____

Touch	Did it fold? (fully / a bit / no)	Time to open again
1		
3		
5		
8		
10		

What I found out

What I noticed:

I think the plant is (learning / tired / something else) because

13 The big argument

Read both sides. Then decide what you think, and why.

Can a plant be intelligent?

A scientist who says yes, in its own way

“Plants sense their world, make choices like which way to grow roots, and change what they do based on what happened before. That's a kind of intelligence, just not ours.”

“We call only brains clever because we have one.”

A scientist who says no

“A thermostat also senses and responds. We don't call it intelligent.”

“Using words like 'learning' and 'feeling' for plants can mislead people. We should describe what they do without guessing what it's like inside.”

I think...

A question I'd ask both of them:

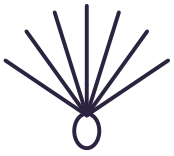
Station 7 **Seeds: travelling and shared** ≈ 45 min ☐ done

14 How do seeds travel?

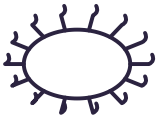
A plant can't walk, so its seeds must travel to find space to grow. Look at each seed's shape. Write each letter in the right column.



A. sal seed with wings



B. aak (madar) seed with silky hairs



C. a hooked burr



D. coconut



E. balsam pod that bursts



F. mango

wind	water	animals	bursting

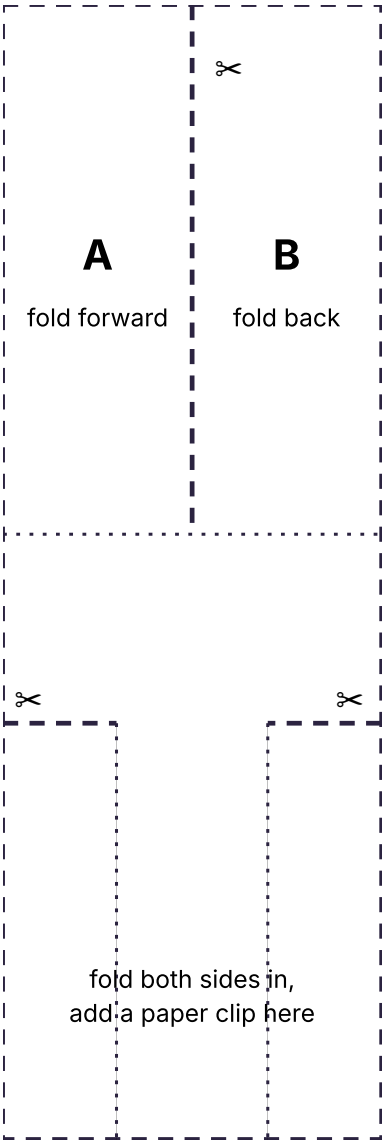
1. A mango tree 'pays' animals with sweet fruit to carry its seeds away. Who is using whom?

_____

_____

15 Make a spinning seed

Some seeds spin like helicopters as they fall, so the wind carries them further. Make one and test it.



1. Cut out the shape along the dashed lines.
2. Cut down the middle of the top part to make two wings.
3. Fold wing A forward and wing B back.
4. Fold the two sides of the bottom part in, and add a paper clip.
5. Drop it from a safe height (a grown-up helps). Time the fall 3 times.
6. Fair test: cut both wings 2 cm shorter and drop it 3 times again from the same place.

Wings	Try 1 (s)	Try 2 (s)	Try 3 (s)
long			
short			

Which fell more slowly? Which would the wind carry further? Why?

Whose seeds are they?

For about 10,000 years, farmers everywhere saved the best seeds from each harvest, planted them again, and swapped them with neighbours. That is how wild grasses slowly became rice and wheat, and it was mostly done by farmers whose names nobody wrote down, many of them women. India once grew tens of thousands of kinds of rice. One scientist, R.H. Richharia, collected about 19,000 varieties from the Chhattisgarh region alone.

Janaki Ammal (1897–1984) was one of India's first women botanists. She bred a sweeter sugarcane for Indian farmers, at a time when women, and people from her community, were often kept out of science. A magnolia flower is named after her. Later in life she worked to save the forests of Silent Valley in Kerala.

In the 1960s, new high-yield seeds helped India grow far more wheat and rice, and saved millions from hunger. But many of these seeds need more water and fertiliser, and some new kinds must be bought again every year. Many old varieties were lost.

16 The seed question

Read both sides. Then decide what you think, and why.

Should anyone be allowed to own a kind of seed?

A seed company scientist says

"It takes years and lots of money to breed a better seed. If we can't sell it, who will pay for the next one?"

"Better seeds feed more people."

A farmer who saves seeds says

"Seeds were bred by farmers for 10,000 years and shared freely. They belong to everyone."

"If I must buy seeds every year, I depend on the company. If I save my own, I am free."

I think…

A question I'd ask both of them:

Station 8 Puzzle room

≈ 30 min ☐ done

17 Who planted what?

Read each clue. Put ✓ for yes and × for no. Every row and column has exactly one ✓.

Four friends each planted one plant on Plant Day: a neem, a rose, mint and a money plant. Use the clues, and what you know about kinds of plants, to find who planted which.

	neem	rose	mint	money plant
Asha				
Bilal				
Chitra				
Dev				

1. Dev's plant will one day be taller than his house.
2. Bilal planted a herb that you can put in chutney.
3. Chitra's plant can't stand up by itself. It needs a stick or a wall to climb.
4. Asha needs gloves when she prunes her plant.

Answer: Asha → _____ · Bilal → _____ · Chitra → _____ · Dev → _____

18 Professor Bholu's notebook

There are 6 mistakes. Underline each one, then write the correct idea.

My notes on plants, by Professor Bholu (very clever).

1. A tree is made mostly from soil. That's why it needs so much.
2. Water travels down from the leaves to the roots.
3. Plants make oxygen, so they don't need any themselves.
4. Roots only need water. They never need air.
5. If you keep a plant in a dark cupboard, it grows extra green.
6. Every plant in the world has flowers.

1. It should say: _____

2. It should say: _____

3. It should say: _____

4. It should say: _____

5. It should say: _____

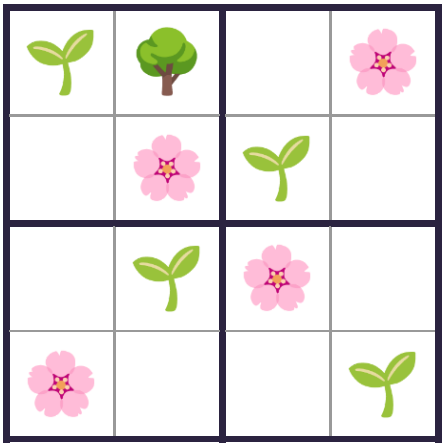
6. It should say: _____

🗝️ **Clue 3** Look again at mistakes 4 and 5 in Professor Bholu's notebook. Do any of them sound like what is happening to Dadi's tulsi?

Copy it onto the Case board on the last pages.

19 Garden sudoku

Every row, every column and every 2×2 box must have each of the 4 pictures once.



The pictures: 🌱 seed 🌿 leaf 🌳 tree 🌸 flower

20 Odd one out

Circle the odd one out and say why. There can be more than one right answer, if you can explain it!

1. mango · coconut · banana · potato

   Odd one: _____ because _____

2. neem · rose · tulsi · mushroom

   Odd one: _____ because _____

3. rice · wheat · bamboo · coriander

   Odd one: _____ because _____

4. cactus · lotus · mangrove · money plant

   Odd one: _____ because _____

21 Riddles

Read each riddle. Write your answer.

1. *I am made mostly of air, yet I can hold up a house. What am I?*

   _____

2. *I drink with my feet and eat with my hands, and my hands are green. What am I?*

   _____

3. *I have no legs, but I travel the world. I wear wings, parachutes or hooks, or I hide inside something sweet. What am I?*

   _____

4. *I am green and I eat sunshine. Without me, nobody on Earth would have breakfast. What am I?*

   _____

22 Secret message from the neem

Use the code to find the hidden message. Write a letter under each picture.

🐞 = A 🔵 = B ⭐ = C 🌸 = E 🔺 = H 🍃 = O 🍎 = R 🐟 = T 🌙 = W

🌙 🌸 🔵 🍎 🌸 🐞 🐟 🔺 🌸 🌸 🐞 ⭐ 🔺 🍃 🐟 🔺 🌸 🍎

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

Station 9 **Solve the case** ≈ 40 min ☐ done

23 Case board

Copy every clue here. Then solve the case.

Clue 1

Clue 2

Clue 3

What is wrong (give 3 reasons):

My plan to fix it:

Day	What I did	How the plant looks
Day 1		
Day 3		
Day 5		
Day 7		
Day 10		
Day 14		

24 The neem question

Read both sides. Then decide what you think, and why.

Remember the story. The city wants to cut 50 old trees to widen a road. Should it?

A shopkeeper says

"The traffic jams are terrible. People waste hours, and ambulances get stuck."

"We can plant new trees somewhere else."

Gauri says

"An old tree gives shade, cools the street, makes oxygen and is home to birds and insects. A sapling takes 20 years to catch up."

"You can only price the wood because the wood is the only part you can sell."

I think...

A question I'd ask both of them:

25 A care card for the tulsi

Draw the healthy tulsi on the left. On the right, write Dadi a care card: where to keep it, how often to water it, what pot to use, and how to tell if it's thirsty.

Station 10 Workshop: make, grow and test

≈ 60 min ☐ done

26 Choose your projects

Choose at least three. Tick the ones you finish.

☐ A. Seed balls (beej gola)

40 min + 2 days drying

You need: seeds of a local tree or vegetables, dry compost, clay or garden mud, water

1. Mix 1 handful of seeds, 3 of compost and 5 of clay.
2. Add a little water and roll balls the size of a laddoo.
3. Dry them in the shade for 2 days.
4. Before the monsoon, throw them onto empty, bare land.
5. Mark where; go back after the rains.

Think: Why wrap the seeds in mud instead of just scattering them?

☐ B. The sweating branch

10 min + 2 hours

You need: a clear plastic bag, a thread, a leafy branch on a plant in the sun

1. Slip the bag over a leafy branch (don't pick it).
2. Tie it loosely at the stem.
3. Leave it in the sun for 2 hours.
4. Look at the inside of the bag.
5. Try a second bag on a branch in the shade.

Think: Where did the water drops come from? Which bag had more?

☐ C. Bubbles from a leaf

10 min + 1 hour

You need: a sprig of hydrilla (aquarium plant) or a fresh spinach leaf, a glass of water, bright sunlight

1. Put the plant in the glass of water.
2. Stand it in bright sunlight.
3. Watch closely, with a hand lens if you have one, every 15 minutes.
4. Move it into the shade. Do the bubbles change?

Think: What gas do you think is in the bubbles?

☐ D. How old is that tree?

20 min

You need: a long string, a measuring tape or ruler, a big tree

1. Wrap the string around the trunk at about the height of a grown-up's chest.
2. Measure the string: that's the girth in cm.
3. Rough rule for many trees: age ≈ girth ÷ 2.5.
4. Measure 3 trees. Ask an elder if they know how old the trees really are.

Think: Why might this rule be wrong for some trees?

☐ E. Paint with plant colours

40 min

You need: haldi, a beetroot, spinach, onion skins, a little soap water or chuna (lime) with a grown-up, paper and brushes

1. With a grown-up, make 4 paints: haldi in water; grated beetroot juice; spinach boiled and mashed; onion skins boiled in water.
2. Paint a picture of the neem tree.
3. Now dab a little soap water on the yellow haldi paint. What happens?

Think: Why do you think the haldi changed colour?

☐ F. Sprout race: light or dark?

10 min + 5 days

You need: moong, two cups with damp cotton, a dark cupboard

1. Put 10 soaked moong on damp cotton in each cup.
2. Put one cup on a sunny windowsill, one in the dark cupboard.
3. Keep both damp. Measure the tallest sprout each day for 5 days.
4. Compare colour and height.

Think: Which grew taller? Which looks healthier? Is taller always better?

Questions nobody has answered (yet)

Science isn't finished

Here are real questions about plants that scientists are still working on. Maybe you will answer one of them.

Flowers appeared on Earth quite suddenly, over 100 million years ago, and spread everywhere. Charles Darwin called this 'an abominable mystery'. Scientists still argue about why it happened so fast.

Do plants feel anything? They sense and respond, but whether there is anything it is like to be a plant, nobody knows, and some say we may never know.

How many kinds of plants are there? Hundreds of thousands are known. Thousands more are probably out there, undiscovered, many in forests that are being cut down before anyone has seen them.

How do trees in a forest affect each other underground? Their roots are joined by threads of fungus. Some scientists think trees share food and warnings this way, like a 'wood-wide web'. Others say the evidence is much weaker than the popular stories. The research goes on.

27 My own question

Write a question about plants that nobody in your family can answer. Then try to find out: ask a gardener, a farmer, a book or a scientist.

- My question:
- Who I asked, and what they said:
- What I think now:

Final challenge

28 Final challenge

No looking back! Then check with the book.

1. **Where does most of a tree's weight come from?**
   ○ the soil ○ carbon dioxide in the air ○ fertiliser
2. **What three things does a leaf need to make food?**

   _____
3. **What gas do we breathe out that plants take in?**
   ○ oxygen ○ carbon dioxide ○ steam
4. **Which way does water travel in a stem?**
   ○ up ○ down ○ it doesn't move
5. **A plant has parallel veins. What kind of roots do you predict?**
   ○ tap root ○ fibrous roots
6. **Why do roots rot in a pot with no hole?**

   _____
7. **Why is 'weed' a tricky word?**

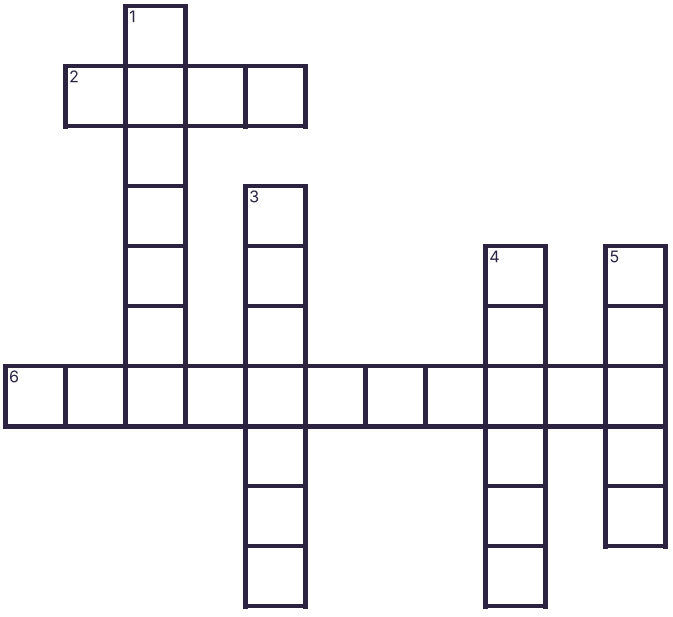
   _____
8. **Name one thing scientists still don't know about plants.**

   _____

My score: ___ / 8

29 Plant words crossword

Read each clue and write the answer in the boxes.



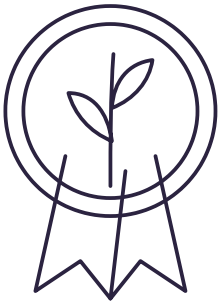
Across →

2 A baby plant packed with food, inside a fruit (4)
6 The green stuff in leaves that catches sunlight (11)

Down ↓

1 A thin curly part a climber uses to hold on (7)
3 Roots that are a bunch of thin ones all the same size (7)
4 The gas leaves give out that we breathe (6)
5 The tiny tubes that carry water up a stem (5)

Certificate



This certifies that

_____

is a

Green Detective

for solving The Case of the Drooping Tulsi, and for thinking like a tree.

_____ _____

Date Signed (a grown-up)

For the grown-up

Guide and answers

How to use this book: start by reading the story aloud together (about 10 minutes); don't explain it, just talk about the questions after it. Then one station a day, 30–40 minutes each, about 6 hours over two weeks. Stations 1 and 2 need waiting, so set them up in the morning.

This book goes far beyond naming parts. The big ideas: a plant builds itself mostly out of air; plants and animals breathe each other; plants sense and respond in ways scientists are still arguing about; 'weed' is a human word; and seeds were shared by farmers for 10,000 years. Let your child disagree with you and with the book. Ask 'How could we find out?' more than you give answers.

The case has a real answer: the tulsi is suffering from three things at once. Too much water with no drainage hole (roots are drowning: they need air, and soggy soil starts to rot, hence the smell), too little light in the kitchen corner (leaves can't make enough food, so they yellow), and the move from balcony to kitchen. Don't give it away; the clues lead there.

Story: Photosynthesis, felt: a tree is built mostly from carbon dioxide taken from the air (plus water), using sunlight; and it gives out the oxygen we breathe. So wood is made of air, and people and trees breathe each other: one cycle. The turn is Gauri's line, 'This tree is made of our breath.' Underneath: told from the tree's point of view; a price that counts only what can be sold; and the women who protect what belongs to everyone (Gaura Devi and the women of Reni, Chipko movement, 1974).

Going further: visit a plant nursery and ask the mali how often they water different plants and why. Visit a botanical garden: the Acharya J.C. Bose Indian Botanic Garden in Howrah, Lalbagh in Bengaluru, or any city garden. Find a sacred grove (devrai, kavu, sarna) near you: patches of forest that villages have protected together for hundreds of years.

How a plant works · Label the plant

1. flower — makes seeds
2. leaf — makes food using sunlight
3. stem — carries water up and holds the leaves to the light
4. root — holds the plant and drinks water from the soil

How a plant works · Mini lab: the covered leaf

The covered part turns paler or yellowish: without light it can't make food or keep making chlorophyll. (Scientists test this properly by checking leaves for starch with iodine.)

Lab: Does water really travel up? · The coloured-water test

The colour climbs up the stem and into the petals or leaf veins: water travels up through tiny tubes in the stem (xylem). You see them as coloured dots or a ring in the cut end.

A tree made of air · Be the scientist

1. No: the soil lost almost nothing, nowhere near 164 pounds.
2. The air (carbon dioxide).
3. Back into the air, as carbon dioxide and water vapour. Only a little ash (the minerals from the soil) is left.

A tree made of air · Breath count

Sitting, children take about 18–25 breaths a minute: roughly 25,000–35,000 a day. About half of the oxygen made each year comes from tiny plants and algae in the sea, the rest from land plants.

Leaf and root detectives · Two kinds of leaf veins

1. Mango, neem, hibiscus, peepal, tulsi, coriander: net (A). Grass, banana, maize, onion, bamboo, wheat: parallel (B).

Leaf and root detectives · Find the secret rule

Net veins usually go with a tap root; parallel veins with fibrous roots. Banana: fibrous. These are two great families of flowering plants (dicots and monocots) that split apart over 100 million years ago. There are exceptions, which is what makes it interesting.

Are plants clever? · Touch-me-not lab
Results vary. A leaf touched many times may fold less or reopen faster. Whether that is 'learning' or tiredness is exactly what scientists argue about.

Seeds: travelling and shared · How do seeds travel?
Wind: A, B. Water: D. Animals: C (hooks), F (eaten, seed dropped far away). Bursting: E. Mango: both use each other; it's a partnership.

Seeds: travelling and shared · Make a spinning seed
Longer wings spin more and fall more slowly, so the wind has more time to carry them away.

Puzzle room · Who planted what?
Asha → rose · Bilal → mint · Chitra → money plant · Dev → neem
Neem is the only tree; mint is the herb; the money plant is the climber; so Asha has the rose, which has thorns.

Puzzle room · Professor Bholu's notebook
1. A tree is made mostly from carbon dioxide in the air (and water); the soil gives only a little in minerals.
2. Water travels UP from the roots to the leaves.
3. Plants need oxygen too: they use it all the time to get energy from their food, day and night, like us. They just make more than they use.
4. Roots need air too; in waterlogged soil they rot.
5. In the dark a plant grows pale and yellow, and stretches tall and thin looking for light.
6. Not all plants have flowers: ferns and mosses don't (they spread by spores).

Puzzle room · Garden sudoku


Puzzle room
1. Potato: it's a swollen underground stem, not a fruit (the others are fruits). Also accept: coconut, the only one that floats across oceans; banana, whose cultivated kind has no real seeds.
2. Mushroom: it isn't a plant at all! It's a fungus: it can't make food from sunlight, and fungi are actually more closely related to animals than to plants.
3. Coriander: rice, wheat and bamboo are all grasses (parallel veins, fibrous roots).
4. Many answers work: money plant (the only climber); lotus (the only one living in fresh water); cactus (built to save water rather than live with lots of it). The reasoning is what counts.

Puzzle room
1. wood (a tree)
2. a plant (roots and leaves)
3. a seed
4. chlorophyll (or a leaf)

Puzzle room · Secret message from the neem
We breathe each other

Solve the case · Case board
1. Too much water and no hole in the pot: the soil stays soaked, the roots get no air and start to rot (that's the smell). 2. Too little light in the kitchen corner: the leaves can't make enough food, so they turn yellow and drop.
3. The move changed its light. Fix: repot into a pot with a hole and a layer of pebbles, water only when the top of the soil is dry, and put it back on the sunny balcony.

Workshop: make, grow and test · Choose your projects
A. Seed balls (beej gola): The clay protects the seeds from birds, ants and the hot sun until the rain softens it; the compost feeds the seedling. Japanese farmer Masanobu Fukuoka made seed balls famous; people in India use them to green bare hills.
B. The sweating branch: Leaves lose water vapour through tiny pores (transpiration), and it condenses on the

bag. More in the sun. This loss is what pulls water up the stem.
C. Bubbles from a leaf: In light, leaves make oxygen during photosynthesis; under water you can see it as tiny bubbles. Fewer bubbles in the shade. (Some bubbles from a spinach leaf are just air that was in the water.)
D. How old is that tree?: It's very rough: fast growers like eucalyptus grow fatter quicker (so they're younger than the rule says), and trees in dry, poor soil grow slowly. Scientists count rings, but that means cutting or drilling, so we estimate instead.
E. Paint with plant colours: Turmeric turns reddish-brown with soap or lime because they are alkaline: it's a natural 'indicator'. Plants are full of chemicals, and people have dyed cloth with them for thousands of years.
F. Sprout race: light or dark?: Sprouts in the dark grow tall, thin and yellow, stretching to find light (Professor Bholu's mistake 5!). The ones in light are shorter, greener and stronger.

Final challenge · Final challenge

1. carbon dioxide in the air
2. sunlight, water and carbon dioxide (air)
3. carbon dioxide
4. up
5. fibrous roots
6. the soil stays soaked, so roots get no air
7. it describes what humans want, not the plant itself
8. e.g. why flowers spread so fast; whether plants feel; how many species exist; how trees affect each other underground

Final challenge · Plant words crossword

